

Azafenidin CAS number 68049-83-2

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Compound of Interest

Compound Name: Azafenidin
CAS No.: 68049-83-2
Cat. No.: B195255

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Azafenidin (CAS Number: 68049-83-2): A Technical Whitepaper

For Researchers, Scientists, and Professionals in Agricultural Chemistry and Environmental Science

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Azafenidin is a herbicide and should be handled with appropriate safety precautions by qualified personnel.

Executive Summary

Azafenidin, a triazolopyridine herbicide, is a potent inhibitor of the enzyme protoporphyrinogen oxidase (PPO), a critical component in the biosynthetic pathway of chlorophyll and heme in plants.[1][2][3] This mode of action leads to the rapid accumulation of protoporphyrinogen IX, which, upon light exposure, causes oxidative damage and subsequent cell death in susceptible weed species.[4][5] This technical guide provides a comprehensive overview of **Azafenidin**, including its chemical and physical properties, mechanism of action, toxicological profile, and available experimental data. The information is presented to support research and professional activities in the fields of agricultural chemistry, weed science, and environmental toxicology.

Chemical and Physical Properties

Azafenidin is a rust-colored solid with a distinct odor.^[1] Its key physicochemical properties are summarized in the table below, providing essential data for experimental design and environmental fate assessment.

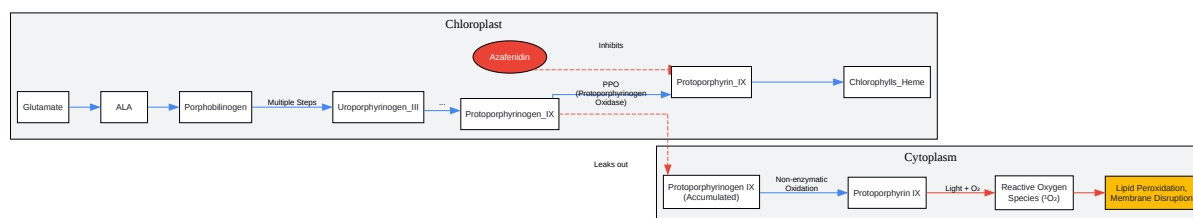
Property	Value	Reference(s)
IUPAC Name	2-[2,4-dichloro-5-(prop-2-yn-1-yloxy)phenyl]-5,6,7,8-tetrahydro[1][6][7]triazolo[4,3-a]pyridin-3(2H)-one	[3][8]
CAS Number	68049-83-2	[1][3]
Molecular Formula	C ₁₅ H ₁₃ Cl ₂ N ₃ O ₂	[1][3]
Molecular Weight	338.19 g/mol	[1][3]
Melting Point	168 - 168.5 °C	[1][9]
Boiling Point	468.9 ± 55.0 °C (Predicted)	[1]
Density	1.4 g/cm ³ (Bulk)	[10]
Vapor Pressure	1 x 10 ⁻⁹ Pa (at 25 °C)	[1]
Water Solubility	12 mg/L (at pH 7)	[1]
Log P (Octanol-Water Partition Coefficient)	2.7	[1]

Mechanism of Action: Protoporphyrinogen Oxidase (PPO) Inhibition

Azafenidin's herbicidal activity stems from its inhibition of protoporphyrinogen oxidase (PPO), the last common enzyme in the synthesis of both chlorophyll and heme.^{[11][12]} PPO is responsible for the oxidation of protoporphyrinogen IX (PPGIX) to protoporphyrin IX (PPIX).^{[2][13]}

The inhibition of PPO by **Azafenidin** leads to a cascade of cytotoxic events:

- Accumulation of Protoporphyrinogen IX (PPGIX): The blockage of PPO causes its substrate, PPGIX, to accumulate within the chloroplast.[4]
 - Extraplastidic Leakage: The excess PPGIX moves from the chloroplast into the cytoplasm.[4]
 - Conversion to Protoporphyrin IX (PPIX): In the cytoplasm, PPGIX is converted to PPIX.[4]
 - Photosensitization and Oxidative Stress: PPIX is a potent photosensitizer. In the presence of light and oxygen, it generates singlet oxygen (1O_2), a highly reactive oxygen species.[5]
 - Cellular Damage: Singlet oxygen causes rapid peroxidation of lipids and proteins, leading to the disruption of cell membranes, leakage of cellular contents, and ultimately, cell death.[2][5]
- This manifests as rapid browning (necrosis) of plant tissues.[2]



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Figure 1: **Azafenidin**'s mechanism of action via PPO inhibition.

Synthesis and Manufacturing

The commercial synthesis of **Azafenidin** presented a significant challenge to meet cost requirements for the specialty-herbicide market.[14] An early laboratory synthesis involved a

classical Fischer hydrazine synthesis followed by coupling and cyclization with phosgene, but this route used hazardous and expensive reagents like butyllithium and phosgene.[14] A key breakthrough for commercial manufacturing was the development of a novel synthesis for the triazolinone ring system and an environmentally benign process for a key intermediate, 5-cyanovaleramide, derived from the selective hydrolysis of adiponitrile.[14]

Toxicological Profile

The toxicological data for **Azafenidin** indicates low acute toxicity to mammals. However, there are concerns regarding reproductive toxicity and potential for organ damage through prolonged or repeated exposure.

Endpoint	Value	Species	Reference(s)
Acute Oral LD ₅₀	>5000 mg/kg	Rat, Mouse	[1][11]
Acute Dermal LD ₅₀	>2000 mg/kg	Rabbit	[1][11]
Acute Inhalation LC ₅₀	>5.3 mg/L	Rat	[1][11]
Aquatic Toxicity (LC ₅₀)	33 mg/L	Rainbow Trout	[1][11]
Aquatic Toxicity (LC ₅₀)	48 mg/L	Bluegill Sunfish	[1][11]
Avian Toxicity (LD ₅₀)	>2500 mg/kg	Bobwhite Quail, Mallard Duck	[1][11]

GHS Hazard Statements:[3]

- H360Df: May damage the unborn child. Suspected of damaging fertility.
- H373: May cause damage to organs through prolonged or repeated exposure.
- H410: Very toxic to aquatic life with long lasting effects.

Environmental Fate

Azafenidin's behavior in the environment is influenced by several factors. It is moderately persistent in soil and its degradation is primarily microbial.[7][15]

Environmental Compartment	Fate and Behavior	Reference(s)
Soil	Moderately persistent. Primary degradation is through microbial pathways. A common degradation reaction is O-dealkylation.	[1] [7] [15]
Water	Stable to hydrolysis. The half-life in water under light is approximately 12 days.	[1]
Photodegradation	Azafenidin is susceptible to photolysis in aqueous solutions. In soil and light, it can also be split to form the triazolinone heterocycle.	[1]
Bioaccumulation	There is a potential risk of bioaccumulation.	[7]

Experimental Protocols

Detailed proprietary experimental protocols are not publicly available. However, a general methodology for assessing the pre-emergence herbicidal efficacy of a compound like **Azafenidin** can be outlined.

Objective: To determine the effective dose range of **Azafenidin** for pre-emergence control of target weed species and to assess crop tolerance.

Materials:

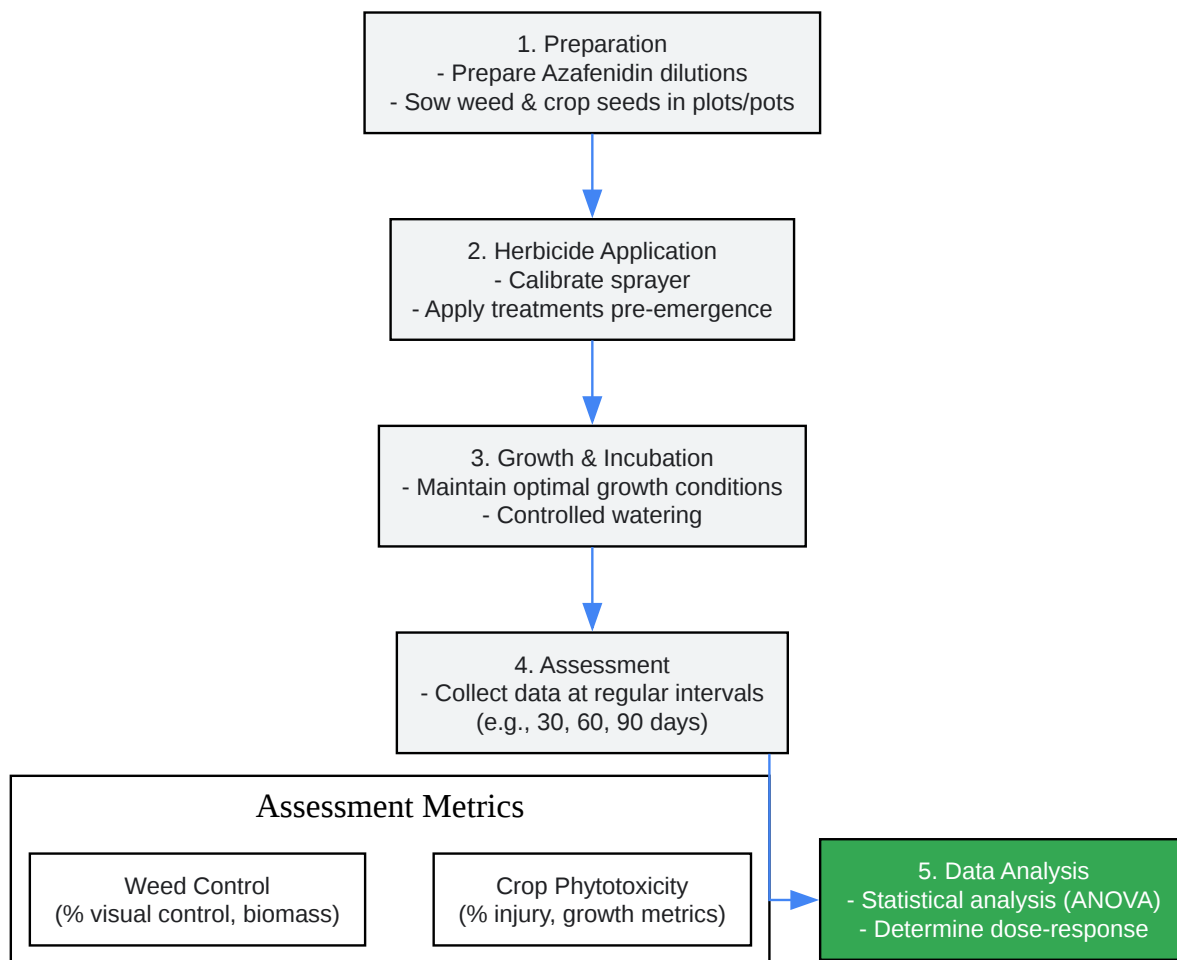
- **Azafenidin** (analytical grade)
- Solvent for stock solution preparation (e.g., acetone with a surfactant)
- Weed seeds (e.g., *Brachiaria decumbens*, *Commelina benghalensis*, *Richardia brasiliensis*)
[\[6\]](#)

- Crop seeds or seedlings (e.g., *Eucalyptus camaldulensis*)[6]
- Pots or field plots with appropriate soil type
- Spraying equipment calibrated for precise application volumes
- Growth chamber or greenhouse with controlled light, temperature, and humidity, or a designated field site.

Methodology:

- Preparation of Test Substance: Prepare a stock solution of **Azafenidin**. Create a series of dilutions to achieve the desired application rates (e.g., 300, 400, 500, 600, 800 g a.i. ha⁻¹).[6]
- Experimental Setup:
 - Fill pots or prepare field plots with soil.
 - Sow seeds of target weed species at a uniform depth.
 - For crop tolerance studies, plant crop seeds or transplant seedlings.
 - Include untreated (weedy check) and weed-free (manually weeded) control groups.[6]
- Herbicide Application:
 - Apply the different rates of **Azafenidin** uniformly over the soil surface before weed emergence (pre-emergence).
 - Ensure consistent application volume across all treatments.
- Incubation and Growth:
 - Maintain the pots or plots under optimal conditions for plant growth.
 - Water the soil as needed to simulate rainfall and activate the herbicide.
- Data Collection and Assessment:

- At regular intervals (e.g., 30, 60, 90, 180 days after application), assess weed control efficacy.^[6] This can be done visually (e.g., % control on a scale of 0-100) or by harvesting and measuring the biomass of surviving weeds.
- Assess crop phytotoxicity by visually rating injury symptoms (e.g., stunting, chlorosis, necrosis) on a percentage scale.^[6]
- Crop growth parameters such as height, stem diameter, and biomass can also be measured.
- Data Analysis:
 - Analyze the data using appropriate statistical methods (e.g., ANOVA, regression analysis) to determine the effective dose for weed control and to assess the significance of any crop injury.



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Figure 2: Workflow for pre-emergence herbicide efficacy testing.

Efficacy Data

Field studies have demonstrated the effectiveness of **Azafenidin** as a pre-emergence herbicide for controlling a range of broadleaf weeds and grasses.

Application Rate (g a.i. ha ⁻¹)	Target Weeds Controlled	Crop Tolerance (Eucalyptus camaldulensis)	Duration of Control	Reference(s)
300 - 400	Good control of most weeds	Minimal phytotoxicity	Shorter residual effect	[6]
≥ 500	Brachiaria decumbens, Commelina benghalensis, Richardia brasiliensis, Sida santaremnensis	Some phytotoxicity observed at ≥ 600 g a.i. ha ⁻¹	Effective up to 180 days after application	[6]
16.0 oz a.i. acre ⁻¹ (~1120 g a.i. ha ⁻¹)	Most key citrus weeds including narrowleaf panicum, Spanish needles, goatweed, pigweed, and balsamapple vine	Excellent tolerance in citrus	4 to 6 months	[15]

Regulatory Status

Azafenidin is not approved for use as a plant protection agent within the European Union.[1][7] An application for its inclusion in Annex I of Directive 91/414/EEC was withdrawn in 2002.[10] Its registration status varies in other parts of the world.

Conclusion

Azafenidin is a highly effective, low-use-rate herbicide with a well-defined mechanism of action targeting the PPO enzyme. Its chemical properties and toxicological profile necessitate careful handling and consideration of its environmental impact, particularly in aquatic systems. The

provided data and protocols offer a foundational resource for researchers and professionals engaged in the study and application of herbicidal compounds. Further research could focus on developing safer analogs, understanding resistance mechanisms, and refining application techniques to minimize environmental exposure.

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